1 | Experiential Learning Component

1.1 | UNIT III : Experiential Learning - 7

[Level-1: 3Q Level-2: 1Q]

1. **LEVEL - 1 : Graph Representation** : Adjacency Matrix Representation - Write a program that represents an undirected graph using an adjacency matrix.
   - Input: Number of vertices and edges, followed by the edges (pairs of vertices).
   - Output: Display the adjacency matrix.
2. **LEVEL - 1 : Graph Representation** : Adjacency List Representation - Write a program to represent an undirected graph using an adjacency list.
   - Input: Number of vertices and edges, followed by the edges (pairs of vertices).
   - Output: Display the adjacency list.
3. **LEVEL - 1 : Problem Solving** : Given a graph with 5 vertices and the following edges: (1,2), (1,3), (2,4), (2,5), represent the graph using an adjacency matrix.
   - Perform BFS traversal Trace in a tabular form starting from vertex 1.
   - Perform DFS traversal Trace in a tabular form starting from vertex 1.
4. **LEVEL - 2 : Graph Traversal** - DFS on Adjacency Matrix - Write a function to perform DFS traversal of a graph represented using an adjacency matrix.:
   - Input: Adjacency matrix, and starting vertex.
   - Output: Display the order of traversal.